

Jal Irani

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RESEARCH INTERESTS	Generative AI, AI agents/agent systems, large language models, retrieval-augmented generation, AI education, blockchain, software engineering, entrepreneurship	
EDUCATION	Dakota State University , Madison, SD Ph.D., Artificial Intelligence, expected: December 2028 Johns Hopkins University , Baltimore, MD M.S., Computer Science, May 2017 Towson University , Towson, MD B.S., Computer Science, May 2015	
HONORS AND AWARDS	Towson University: Outstanding Lecturer Award Technically: Top 21 Most Influential Engineers in Baltimore University System of Maryland: COVID App Challenge Winner Mary Hudson Scarborough Award: Top Graduating Computer Science Student	2024 2021 2020 2015
ACADEMIC EXPERIENCE	Towson University Towson, MD <i>Associate Teaching Professor</i> August 2024–Present <i>Lecturer</i> August 2017–July 2024 Course lead for the department's blockchain, mobile application development, and web development curriculum. Relaunched the department's Industry Advisory Board (IAB), recruiting and coordinating 10+ industry leaders to advise curriculum and strategic program direction. Developed new undergraduate and graduate courses in blockchain, mobile development, and emerging AI technologies, including launching COSC 141, the department's first Generative AI course for non-majors. <i>Assistant Director, AI Institute</i> January 2026–Present Founded and launched Towson University's first AI Clinic, partnering with startups and small businesses to develop AI strategy, identify implementation opportunities, and deploy student consulting teams as forward-deployed AI engineers. Leads faculty development initiatives on modern AI technologies, including training TSEM instructors who teach incoming first-year students. Collaborates with university leadership to develop and implement Towson's institution-wide AI strategy. Johns Hopkins University Baltimore, MD <i>Adjunct Professor</i> August 2019–Present Teaches graduate courses in both the Carey Business School and the Whiting School of Engineering, spanning machine learning, generative AI, blockchain, and entrepreneurship. Serves as course lead for Generative AI for Entrepreneurs within the Whiting School of Engineering, overseeing curriculum development and continuous course modernization. Redesigns new graduate courses in generative AI and AI for business, translating emerging AI technologies into practical applications for technical and business professionals.	

Loyola University Maryland

Baltimore, MD

*Adjunct Professor***May 2023–December 2023**

Taught undergraduate computer science courses in mobile application development and blockchain within the Department of Computer Science. Continues to serve on the department's Industry Advisory Board, advising on curriculum development and alignment with emerging industry trends.

GRANTS

Maryland Industrial Partnerships (University of Maryland)	2026
Principal Investigator, \$100,000	
Baltimore Innovation Initiative (TEDCO)	2025
Co-Principal Investigator, \$30,000	
Innovation & Entrepreneurship Award (Johns Hopkins University)	2025
Principal Investigator, \$10,000	
RISE (Comcast)	2024
Principal Investigator, \$20,000	
Energizing Small Businesses (BG&E)	2024
Principal Investigator, \$20,000	
GenCyber (National Security Agency)	2023
Senior Personnel, \$2,000,000	
Urban Business Innovation Initiative (TEDCO)	2023
Principal Investigator, \$10,000	
Maryland Business Innovation Association (TEDCO)	2022
Principal Investigator, \$10,000	
RUBRIC (TEDCO)	2021
Principal Investigator, \$100,000	
I-Corps (Johns Hopkins University)	2020
Principal Investigator, \$2,880	

TEACHING**Towson University**

COSC 670: Cryptocurrencies & Blockchain	2022–2024
COSC 617: Advanced Web Development	2017–2026
AIT 670: Generative AI for Entrepreneurs	2026
AIT 618: Client/Server Programming	2017–2026
AIT 616: Fundamentals of Web Programming	2018–2026
COSC 486: Cryptocurrencies & Blockchain	2022–2026
COSC 484: Web Programming	2017–2026
COSC 435: Mobile Development	2024, 2026
COSC 141: AI Technology in Practice	2026
COSC 109: Computers & Creativity	2024–2026

Johns Hopkins University

EN.635.674: Generative AI for Entrepreneurs	2024–2026
BU 520.775: Practical Machine Learning	2025
Created from BU 520.710	
BU 520.710: AI Essentials for Business	2025, 2026
Created from BU 520.710	
BU 520.710: Big Data Machine Learning	2021, 2023, 2024
Course later split into BU 520.775 and BU 520.710	
BU 330.760: Generative AI	2026
BU 232.770: Cryptos & Blockchain	2019, 2020, 2021, 2025

Loyola University Maryland

CS 489: Cryptos & Blockchain

2023

CS 453: Mobile Development

2023

PROFESSIONAL EXPERIENCE

Coursta

Baltimore, MD

Founder & CEO

August 2026–Present

Founded an AI-powered degree planning platform that helps universities streamline curriculum mapping, degree audits, and student course planning using large language models and agentic AI.

GoPanda

Baltimore, MD

Founder & CEO

January 2025–Present

Founded and scaled a software development studio delivering AI and web solutions for startups; built partnerships with venture-backed clients and advised founders on MVP development, product-market fit, and fundraising strategy.

Featured Partners: GalaxyBids, StorytimeAI, BettorVision, getPERKd, MyHealthData

Flave (Acquired: Rallie)

Baltimore, MD

Founder & CEO

January 2020–June 2026

Built and launched a local dining rewards platform from concept to market leader; managed fundraising, team hiring, and partnerships, securing municipal and media recognition.

Featured: Collegepark.gov, Company to Watch (2023), Top 20 Startup of 2022, WBAL, BBJ, Technical.ly

clean.io (Acquired: Human)

Baltimore, MD

Head of Product

January 2020–February 2021

Led the vision, strategy, and execution for a suite of digital engagement security products (cleanAD and cleanCART), while growing revenue 175% in one year.

CountySportsZone

Baltimore, MD

Chief Technology Officer

November 2018–January 2020

Led the vision, strategy, and execution for a suite of digital engagement security products (cleanAD and cleanCART), while growing revenue 175% in one year.

TrackOFF (Acquired: Avast)

Baltimore, MD

Lead iOS Developer

November 2017–November 2018

Supported and worked on all phases of software project delivery, including initial client engagement and requirements gathering, user centered design, agile development practice, CI/CD, hand-off and knowledge transfer to the client, and post-production support and trouble-shooting.

ZeroFox (Acquired: Haveli)

Baltimore, MD

Full-Stack Developer

January 2016–November 2017

Created asynchronous rules using JavaScript to alert the ZeroFOX platform on threatening/suspicious activity using social media APIs. Designed and taught a two-day JavaScript-based Security Engineer training curriculum for non-technical customers.

Mindgrub

Baltimore, MD

Software Engineering Intern

September 2015–December 2015

Wrote automated UI tests using Calabash to ensure new features and version updates did not affect any use cases of the Polk Audio Omni Utility application.

Primary Data

Los Altos, CA

iOS Developer Intern

May 2015–August 2015

Developed a proprietary iOS application to visually define memory management options for a

software-defined storage system. The resulting application was presented at VMworld 2015 in SF, California.

Towson University

Towson, MD

iOS Developer

August 2014–May 2015

Maintained and redeployed the Wavyleaf application to allow users to plot, photograph, and detail their sightings of the invasive plant species.

Textron Systems

Hunt Valley, MD

Software Engineering Intern

May 2014–August 2014

Modified testing tools to simulate the embedded C++ airborne software system to support multiple payload configurations.

Vane Brothers

Baltimore, MD

Software Engineering Intern

May 2013–August 2013

Wrote a proprietary C# application that tracks the location, speed, direction, weather, fuel, payload, and crew information for the entire 120 vessel fleet.

CONSULTING

Towson StarTUP

2026

Architected and launched the digital visitor check-in platform for the StarTUP at the Armory, which has been used by over 15,000 unique visitors.

Fourplay

2023

Led development of the mobile application from concept through public launch, surpassing 100,000 downloads and peaking at #89 on the Apple App Store.

Featured: CNBC, New York Post

Corrie

2016

Led the development and launch of the Corrie mobile application as part of a collaboration between Apple and the Johns Hopkins Hospital.

Featured: Apple WWDC 2021

PROJECTS

SleepIn (Acquired: App Masters)

2018–2024

Built a school cancellation alarm application that automatically delayed users' alarms during snow days and weather closures.

Featured: WBAL, The Diamondback, Baltimore Watchdog

SmartAddress (Acquired: FamJam)

2016–2018

Built an iOS application that compares Uber and Lyft fares in real time.

PROFESSIONAL
SERVICE

StarTUP Accelerator

2020–Present

Mentor

University of Maryland Blockchain Accelerator

2022–Present

Mentor

Loyola University Maryland, Department of Computer Science

2023–Present

Industry Advisory Board Member

New York City Department of Education (NYC DOE)

2025–Present

External Review Committee Member

Business Analytics Artificial Intelligence Organization (BAAIO)

2026–Present

Faculty Advisor

TapIn

2022–2024

Advisory Board Member

eTU Council

2022–2024

Entrepreneurship Faculty Fellow

CountySportsZone

2018–2020

Advisory Board Member

SKILLS

Programming Languages: (order of proficiency) Swift, JavaScript (ES12+), Python, Java, Solidity, Rust

Agentic AI: OpenAI Agents SDK, Anthropic Agent SDK, LangGraph, Agent-to-Agent (A2A), Model Context Protocol (MCP), Tool Calling, Multi-Agent Orchestration

Generative AI: Retrieval-Augmented Generation (RAG), LLM Fine-Tuning, Vector Databases (FAISS, Pinecone)

Machine Learning: PyTorch, TensorFlow, scikit-learn

iOS Development: SwiftUI, UIKit

Web Development: React.js, Node.js, Express.js

Web3: IPFS, Alchemy, Candy Machine